

Phase 7 Aggregate Report

Overview

- Run count: 20.
- Condition count: 7.
- Best transfer condition: phase7_full_solver_evolution.

Condition Means

Condition	Final Transfer Gap	Final TSPLIB Gap	Final Family Gap	Mean Novelty	Mean Complexity	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	0.121317	0.235058	0.054968	0.0	0.0	0.0	0.0
phase7_full_solver_evolution	0.117916	0.164668	0.036098	0.832777	0.65	0.0	0.0
phase7_modular_operator_compression_pressure	0.118865	0.232947	0.052316	0.560556	0.44375	0.005698	0.802631
phase7_modular_operator_diversity_residual_replay	0.121843	0.231468	0.057894	0.711483	0.443	0.003445	1.684384
phase7_modular_operator_evolution	0.119602	0.232023	0.054022	0.804885	0.442875	0.003951	1.801854
phase7_modular_operator_pareto_selection	0.126012	0.237085	0.06122	0.591492	0.44475	0.006265	0.196727
phase7_modular_operator_random_replay	0.118833	0.230474	0.053709	0.771204	0.4445	0.003847	0.476892

Paired Comparisons

- modular_vs_baseline_only: transfer delta -0.001715 (95% CI -0.003418 to 8.5e-05), TSPLIB delta -0.003035.
- modular_vs_full_solver: transfer delta 0.001686 (95% CI -0.018321 to 0.022827), TSPLIB delta 0.067355.
- random_vs_modular_base: transfer delta -0.000768 (95% CI -0.002519 to 0.000889), TSPLIB delta -0.001549.
- diversity_residual_vs_random: transfer delta 0.003009 (95% CI -0.002884 to 0.010797), TSPLIB delta 0.000994.
- compression_vs_modular_base: transfer delta -0.000737 (95% CI -0.004573 to 0.002479), TSPLIB delta 0.000924.
- pareto_vs_modular_base: transfer delta 0.006411 (95% CI 0.001983 to 0.012533), TSPLIB delta 0.005061.

Rediscovery

- Signature count: 56.
- Most common surviving signature: None.

Discovery Answers

- modular_vs_full_solver: Relative to full-solver evolution, the base modular operator condition changed transfer by 0.001686 and transplant quality by 0.003951.
- pareto_tradeoff: Pareto selection changed transfer by 0.006411 and runtime inflation by -1.605126 relative to the base modular condition.
- rediscovery: Rediscovery produced 56 distinct signatures; the most common surviving signature is None.
- replay_mechanism: Random replay changed modular transfer by -0.000768 versus the no-replay modular baseline, while diversity-residual replay changed transfer by 0.003009 versus random replay.